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CLASS: BSAI **SECTION:3A**

Lab task:1

Task no:1

History of AI from 1950 to 2017.

The timeline below outlines key events within the development of AI. This is a historical evidence task.

Year/Timeframe

Event

Significance

Source

1950

Alan Turing proposes the Turing Test:

Turing introduces the idea of the imitation game to discuss if a machine could imitate intelligent behaviour in his Computing Machinery and Intelligence paper.

1956

Dartmouth Summer Research Project on Artificial Intelligence

The Dartmouth conference helped create AI as its own field of research. The name Artificial Intelligence became associated with this field.

1966

ELIZA chatbot:

ELIZA, a chatbot developed by Joseph Weizenbaum, utilised pattern recognition techniques to mimic dialogue. This showed early examples of natural language interaction between humans and computers.

1974-1980

AI Winter 1:

AI suffered from criticisms and poor outcomes due to initial promises being too much for the technology to deliver.

1997

IBM Deep Blue beats Garry Kasparov:

Deep Blue won against the current world chess champion. This showed how specific algorithms along with high computing power could surpass the best human in a complex task.

2012

Alex Net ImageNet:

Alex Net was able to achieve a huge result in ImageNet. This showed how having large image data sets and modern computing power can aid deep learning. This used a deep convolutional neural network and efficient GPU implementation.

2017

Transformer Architecture:

In the paper ‘Attention Is All You Need’, the Transformer architecture was introduced. This used the attention mechanism and was more parallelisable. This became fundamental in many modern language models.

Task no:2

```
def check_result(score, documents, attendance, verification):
if score >= 70 and documents == True and attendance >= 75 and verification >= 60:
return "Pass"
else:
return "Fail"
test_cases = [
(85, True, 80, 75),
(85, True, 70, 75),
(85, True, 80, 55),
(70, True, 75, 60)
]
for i in range(4):
score = test_cases[i][0]
documents = test_cases[i][1]
attendance = test_cases[i][2]
verification = test_cases[i][3]
actual = check_result(score, documents, attendance, verification)
```

```
if i == 0:
expected = "Pass"
reason = "All conditions are satisfied."
elif i == 1:
expected = "Fail"
reason = "Attendance is below 75%."
elif i == 2:
expected = "Fail"
reason = "Verification score is below 60."
else:
expected = "Pass"
reason = "All values are at the threshold."
print("Test Case:", i + 1)
print("Input:", test_cases[i])
print("Expected Result:", expected)
print("Actual Result:", actual)
print("Reason:", reason)
print("----------------------")
```

Task no: 3

Conversational Test and General Intelligence

In the scenario of applying for a scholarship, a conversational test could be used to see if the AI can talk to the student, answer questions on whether they are eligible and explain the requirements for applying using natural language. However, passing the conversational test doesn't guarantee that the AI has general intelligence.

A conversational test will mainly look at the AI's ability to reply naturally. However, it won't provide evidence that the AI fully understands situations, checks information correctly and makes reliable decisions. The imitation game was created by Alan Turing in 1950 to discuss machine intelligence through conversations. However, the AI having good conversational skills doesn't mean it has general intelligence.

Example of Flawed Reasoning

If a student asked the AI:

"Can I get a scholarship with 68 marks when the minimum requirement is 70?"

The AI could reply:

"Yes! You can get the scholarship because your academic results are good and your full documents show you're a serious student!"

This seems very friendly and trustworthy however, this isn't correct as the student hasn't reached the 70 required marks.

Better Test: Accurate Work

The AI could be asked to look at 4 different scholarship applications, all with different scores and documents. They would need to work out whether each application would qualify based on the real requirements of the scholarship. We can compare their answers to the correct answers and see if they have given accurate reasoning for why they passed or failed.

Conclusion:

A conversational test will show how natural the AI can communicate however, a work test will show whether they can do work accurately and consistently. Just because the AI could pass a conversational test doesn't mean they have general intelligence.

References:

Turing, A. M. (1950). Computing Machinery and Intelligence. [online] Oxford Academic.

Searle, J. R. (1980). Minds, Brains, and Programs. [online] Behavioral and Brain Sciences

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GITHUB LINK:

https://github.com/abdullahrehman957-lgtm/AI-LAB-TASK
